

Current Project Directory

None

```
/home/aniruddhan/  
|  
├─ tcu_1publisher_mqttCAN  
├─ tcu_1publisher_mqttCAN.cpp  
|  
├─ vehicle_logger.py  
|  
├─ dashboard.py  
|  
└─ vehicle_telematics_v2.db
```

File 1: Publisher

None

```
tcu_1publisher_mqttCAN
```

Compiled executable.

Generated from:

None

```
g++ tcu_1publisher_mqttCAN.cpp \  
-o tcu_1publisher_mqttCAN \  
-lmosquitto \  
-std=c++17
```

Run:

None

```
./tcu_1publisher_mqttCAN
```

File 2: Source Code

None

`tcu_1publisher_mqttCAN.cpp`

Not needed at runtime if executable exists.

Keep for:

- GitHub
- Resume
- Future modifications

File 3: MQTT → SQLite Logger

None

`vehicle_logger.py`

This:

None

`MQTT`

↓

`SQLite`

subscribes to:

None

`vehicle/telematics`

and writes to:

None

`vehicle telematics_v2.db`

Run:

None

`python3 vehicle_logger.py`

File 4: SQLite Database

None

`vehicle telematics_v2.db`

Created automatically by:

None

`vehicle_logger.py`

Used by:

None

`dashboard.py`

File 5: Dashboard

None

`dashboard.py`

Reads:

None

```
vehicle telematics_v2.db
```

and displays:

- Gauges
- Maps
- Charts
- Analytics

Run:

None

```
streamlit run dashboard.py
```

Runtime Dependency

Your dashboard contains:

None

```
conn = sqlite3.connect(  
    "vehicle telematics_v2.db"  
)
```

Therefore:

None

```
dashboard.py
```

and

None

```
vehicle telematics_v2.db
```

must either:

Option A

Be in the same directory

```
None  
dashboard.py  
vehicle_telematics_v2.db
```

or

Option B

Use a full path:

```
None  
conn = sqlite3.connect(  
    "/home/aniruddhan/database/vehicle_telematics_v2.db"  
)
```

Required Running Terminals

Terminal 1

Publisher

```
None  
./tcu_1publisher_mqttCAN
```

Terminal 2

Logger

```
None  
python3 vehicle_logger.py
```

Terminal 3

Dashboard

None

```
streamlit run dashboard.py
```

Mosquitto Service

Must be running:

Check:

None

```
sudo systemctl status mosquitto
```

Start:

None

```
sudo systemctl start mosquitto
```

Enable at boot:

None

```
sudo systemctl enable mosquitto
```